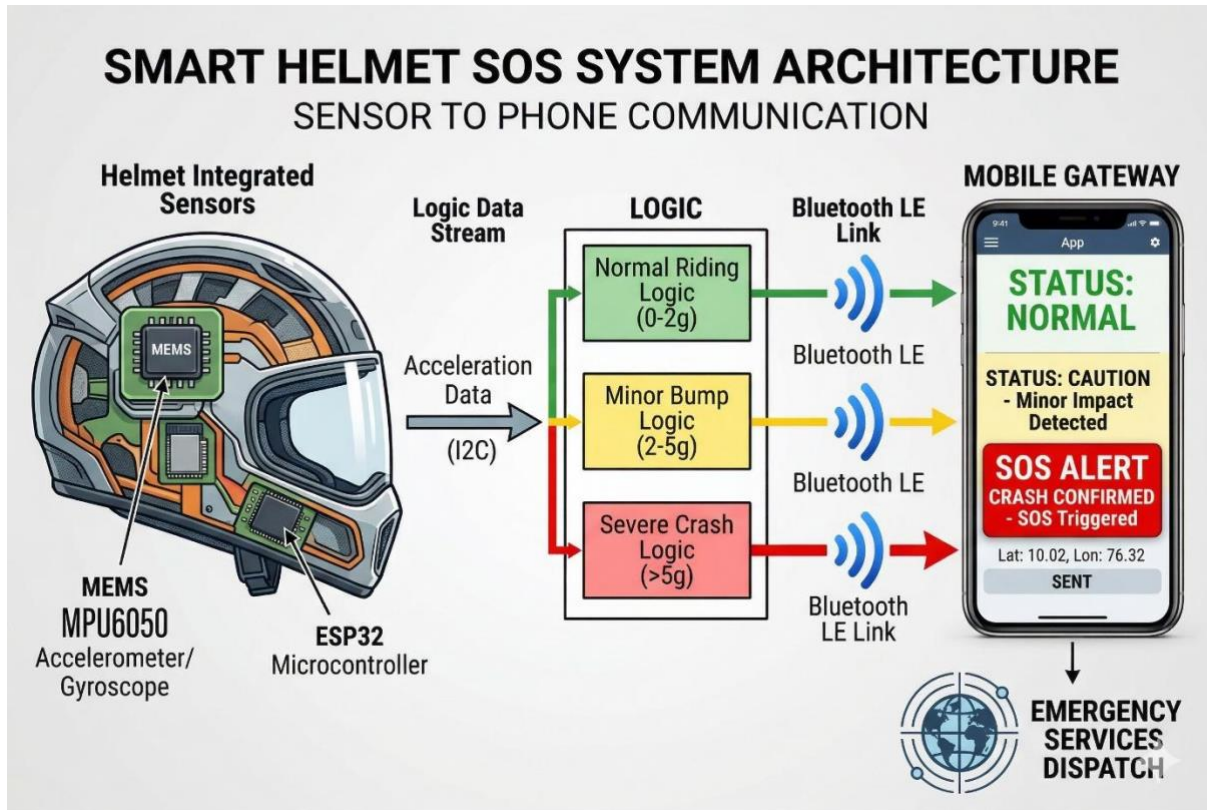




CERTIFICATE



This is to certify that the report entitled **HELMET SOS SYSTEM** submitted by **ALAN LIJO (PKD2SEC006), AMITH KISHAN P P (PKD2SEC009), MUHAMMED SHAFADH T K (PKD2SEC045), SHRISH M (PKD2SEC059)** to the APJ Abdul Kalam Technological University in partial fulfillment of the B.Tech. degree in Electronics and Communication Engineering is a bonafide record of the Project Phase II carried out by them under our guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

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DECLARATION

I, the undersigned, hereby declare that the project report entitled “Smart Helmet SOS System using MPU6050 and Bluetooth Technology” submitted to the Department of ECE, Govt. Engineering College Palakkad, in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in Electronics and Communication Engineering under APJ Abdul Kalam Technological University, is a record of original work carried out by me under the guidance of Prof. Soumya C.P.

I further declare that the results embodied in this project have not been submitted to any other University or Institute for the award of any degree or diploma.

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ABSTRACT

The rapid increase in the number of two-wheelers on the road has led to a significant rise in traffic accidents, many of which prove fatal due to delayed medical assistance. This project presents the design and development of a Smart Helmet SOS System aimed at enhancing rider safety through automated accident detection and emergency communication.

The system utilizes an MPU6050 MEMS sensor, incorporating a 3-axis accelerometer and a 3-axis gyroscope, to monitor the helmet's orientation and detect sudden impacts or abnormal tilt angles indicative of a crash. Data processing is handled by a microcontroller, which analyzes the sensor values in real-time. Upon detecting a potential accident, the system triggers a communication module via Bluetooth to interface with a dedicated mobile application. The smartphone then automatically sends an SOS alert, including the rider's geographical coordinates, to pre-defined emergency contacts.

By bridging the critical gap between the occurrence of an accident and the arrival of emergency services, this system provides a low-cost, reliable solution for real-time monitoring and life-saving intervention. Experimental results demonstrate that the system effectively distinguishes between routine helmet movements and actual impact events, ensuring high reliability and minimal false alarms.

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1. Introduction: Smart Helmet SOS System

Road safety for two-wheeler riders remains a critical global concern. While standard helmets offer physical protection, many fatalities occur because emergency services are not notified in time—a period known as the "Golden Hour." The **Smart Helmet SOS System** addresses this by transforming traditional safety gear into an active, intelligent communication hub that detects accidents and summons help automatically.

Project Overview and Mechanism

The system is built around a low-power microcontroller (such as the **ESP32** or **Arduino**) integrated with an **MPU6050** sensor. This sensor utilizes a 3-axis accelerometer and a 3-axis gyroscope to monitor the helmet's motion and orientation in real-time. By processing g -force data and angular velocity, the system can distinguish between normal riding movements and a high-impact collision.

To ensure reliability, the firmware employs a threshold-based algorithm. If an impact exceeds a specific limit, the microcontroller confirms the accident and triggers the communication sequence. This eliminates the dependency on the rider—who may be unconscious—to call for help manually.

Connectivity and Impact

Once an accident is detected, the system uses **Bluetooth** technology to pair with the rider's smartphone. A dedicated application then retrieves the GPS coordinates of the crash site and sends an automated SOS message to emergency contacts and local medical services.

Key Objectives:

- **Immediate Notification:** Dramatically reduces the time between an accident and the arrival of medical aid.
- **Automated Accuracy:** Provides precise GPS locations to responders, minimizing search time in remote areas.
- **Scalability:** The framework allows for future additions like alcohol detection or "No Helmet, No Start" ignition interlocks.

In conclusion, this project leverages embedded systems and sensor fusion to bridge the gap between accident occurrence and medical response. By integrating smart technology into everyday safety equipment, the **Smart Helmet SOS System** provides a proactive solution to improve survival rates on the road.

2. Problem Statement: The Crisis of Delayed Emergency Response

Two-wheeler accidents account for a disproportionately high percentage of road fatalities globally. While a helmet is the most effective tool for preventing traumatic brain injuries, its utility ends at physical protection. The primary factor determining the survival of a rider after a high-impact collision is the **speed of medical intervention**.

The current landscape of road safety faces three critical challenges:

The "Golden Hour" Gap: Most fatalities occur because the victim does not receive medical attention within the first hour of the accident. In rural or isolated areas, or during late-night hours, a rider may lie incapacitated for an extended period before being discovered by passersby.

Rider Incapacitation: In severe accidents, the rider is often rendered unconscious or is in a state of shock, making it physically impossible for them to reach for a phone to dial emergency services or share their location.

Lack of Real-Time Information: Emergency responders often struggle to locate accident sites quickly due to vague reports from witnesses. Without precise GPS coordinates, valuable minutes are lost navigating to the scene.

The Need for an Automated Solution

Current safety measures are **passive**—they only react to the physical impact. There is a glaring lack of **active** safety systems that can autonomously bridge the communication gap between the accident site and emergency services.

The **Smart Helmet SOS System** is proposed to solve this problem by removing human intervention from the emergency alert process. By utilizing embedded sensors and wireless connectivity, the system aims to automate the "cry for help," ensuring that even if a rider is alone and unconscious, their precise location and status are communicated to responders instantly.

3. Objectives of the Project - “Smart Helmet SOS System”

1. To design an intelligent safety system that detects road accidents in real-time using an MPU6050 motion sensor.
2. To establish a seamless wireless communication link between the helmet and a smartphone using Bluetooth technology.
3. To automate the process of sending SOS alerts with precise GPS coordinates to emergency contacts without manual intervention.
4. To implement a robust algorithm that distinguishes between actual high-impact collisions and minor incidents to prevent false alarms.
5. To apply core principles of Embedded Systems, C programming, and Sensor Fusion in a practical safety application.
6. To fabricate a compact and energy-efficient prototype that can be easily integrated into existing helmet designs.
7. To reduce the critical emergency response time (the "Golden Hour") by ensuring immediate notification of medical authorities.
8. To evaluate the performance and sensitivity of the accelerometer-based trigger under various simulated impact conditions.
9. To develop comprehensive technical documentation including circuit schematics, firmware logic, and testing results as per project requirements.

4. Scope of the Project

The project “**Smart Helmet SOS System**” is intended primarily for educational demonstration, safety training, and light-duty personal transportation safety applications. The scope is deliberately limited to ensure simplicity, safety, and feasibility within the framework of an engineering micro project.

- From an academic perspective, the project serves as a practical learning model that helps students apply fundamental concepts from **Embedded Systems, Programming, and Sensor Fusion**.
- The automated SOS alert system makes it easier to understand the relationship between physical impact forces and wireless digital response, making the prototype suitable for classroom and laboratory demonstrations.
- The project also has relevance in road safety awareness environments, such as training centers, where two-wheeler safety is a priority.
- The integrated warning system promotes awareness of the "Golden Hour" in medical emergencies and encourages responsible riding practices without relying on complex, factory-installed vehicle telematics.
- The application of the project is limited to prototype testing and moderate impact scenarios suitable for demonstration purposes.
- It is not designed for heavy industrial use, certified forensic accident analysis, or as a replacement for professional emergency medical dispatch systems.
- Industrial safety standards, high-durability hardware requirements, and long-term battery performance in extreme weather are beyond the scope of this project.
- Overall, the defined scope ensures that the project remains educationally valuable, safe to operate, and achievable using institutional resources, while clearly acknowledging its limitations in industrial and certified applications.

5. Literature Review / Existing Methods

Two-wheeler safety devices, such as standard helmets and basic protective gear, are widely used in personal transportation, workshops, and delivery services due to their simplicity, portability, and low cost. These conventional systems rely entirely on the helmet's physical structure to protect the rider and typically consist of basic mechanical components like an outer shell, impact-absorbing liner, and chin straps. While these devices are effective for protecting the head within rated impact capacities, they do not provide any real-time feedback or visual/auditory indication of an accident magnitude. Operators must rely on external help or their own consciousness to report a crash, which can lead to unsafe conditions and fatalities during the "Golden Hour".

Some advanced vehicle safety systems, such as high-end motorcycles and luxury cars, incorporate digital crash sensors, e-call features, and satellite-linked emergency displays to show real-time accident data. These systems provide accurate impact measurements and automatic safety alarms when a collision is detected. However, such systems are expensive, require complex vehicle integration, and involve sophisticated sensor calibration and subscription-based cellular services. As a result, they are typically found in premium automotive segments where cost and infrastructure support justify the added complexity. They are not practical for the average commuter or training environments where standard manual safety equipment is preferred for its costeffectiveness.

A few aftermarket devices and standalone safety trackers have been developed, such as mechanical impact stickers, GPS pucks, and wearable SOS buttons used in adventure sports and lone-worker monitoring. Although these devices can detect or report a fall, most are standalone instruments not integrated with the helmet's core protective function. They either require separate charging, do not fit easily within the helmet's ergonomics, or lack clear automated signaling for emergency responders.

In conclusion, existing helmets lack built-in accident communication, and existing electronic safety tools are either too expensive or unsuitable for seamless integration with standard protective gear. This gap in current systems demonstrates the need for a low-cost automated SOS indicator tailored specifically to two-wheeler safety applications, which is the prime focus of this micro project.

6. Market Need & Gap Analysis

Individual commuters, delivery services, and two-wheeler training institutes form a significant segment of the transportation and logistics ecosystem. These environments frequently rely on standard helmets and basic protective gear because of their low cost, ease of use, and minimal maintenance requirements. Unlike high-end automotive sectors, such establishments and individuals often operate with limited budgets and cannot afford advanced motorcycles with built-in electronic safety features or subscription-based emergency calling.

Despite their widespread use, standard helmets used in these settings lack basic safety aids, particularly systems that indicate or report an accident automatically. Riders and emergency responders are required to rely on witnesses or the victim's consciousness to report a crash, which is often impossible in severe accidents. This limitation becomes more critical in remote areas or late-night travel where riders may be inexperienced or alone and unaware of the "Golden Hour" medical window. As a result, delayed medical intervention may occur, leading to unsafe conditions, permanent disability, and potential fatalities.

Although advanced vehicle safety systems provide digital crash displays and automated e-calls, these solutions are expensive, power-dependent, and complex, making them unsuitable for small-scale personal applications. On the other hand, existing helmets continue to be sold without any form of automated SOS indication. This creates a clear market gap between high-cost automotive safety systems and low-cost personal protective equipment.

The proposed **Smart Helmet SOS System** directly addresses this gap by offering a simple, automated, and affordable safety solution. By using intuitive Bluetooth-linked alerts to represent impact and emergency conditions, the system enhances safety awareness without adding significant cost or complexity. The solution is particularly well-suited for daily commuters and delivery fleets where simplicity, reliability, and affordability are critical requirements. The project fulfills an unmet need by introducing an economical safety enhancement to conventional helmets, making it both technically relevant and socially beneficial.

7. Innovation and Value Proposition

The key innovation in the project “**Smart Helmet SOS System**” lies in the conversion of physical impact data into an automated emergency communication sequence without the use of complex vehicle-integrated telematics or expensive subscription services. Unlike conventional road safety monitoring systems that depend on high-end motorcycle sensors, internal power grids, and digital dashboards, the proposed system employs a compact embedded arrangement to represent accident conditions in an intuitive and automated manner.

The design uses fundamental engineering principles such as acceleration measurement, sensor fusion, and wireless data transmission to translate a physical collision into a digital SOS trigger. This trigger is directly linked to a smartphone application that initiates an emergency alert containing precise GPS coordinates. The use of automated messaging as the primary mode of indication ensures instant notification, even if the rider is incapacitated. This simplicity significantly reduces the chances of missing the "Golden Hour" and improves operational safety.

A major advantage of the proposed system is its ease of use. Since the indicator does not require complex vehicle wiring, specialized calibration software, or skilled maintenance, it can be easily installed and operated by individual commuters and delivery services. The integration of an independent microcontroller also increases the reliability of the system by eliminating potential issues related to vehicle battery failure or signal errors from the motorcycle's internal bus.

On economic perspective, the innovation emphasizes affordability. The system is fabricated using low-cost, readily available components such as the **MPU6050** sensor, an **ESP32** or **Arduino** microcontroller, and standard Bluetooth modules. This makes the solution accessible to everyday riders who cannot invest in expensive automotive safety equipment. Despite its simplicity, the SOS system provides significant safety benefits by alerting emergency contacts before critical time is lost. Overall, the project demonstrates how a simple embedded innovation can effectively enhance safety, usability, and cost efficiency, making it a practical and impactful solution for two-wheeler safety applications.

8. Intellectual Property Rights

Intellectual Property Rights (IPR) play an important role in protecting innovative ideas and encouraging technological development. In the micro project “**Smart Helmet SOS System,**” the core innovation lies in the conceptual design and working mechanism that converts physical impact data into a visible and automated SOS alert. Although the project is primarily developed for academic purposes, the concept has potential for further development and commercialization, making IPR considerations relevant.

If the design is commercialized, the innovative aspects of the mechanism—such as the specific algorithm for accident confirmation and the integrated sensor-to-mobile communication loop—may be protected through patent filing. A patent would grant exclusive rights to the inventor to manufacture, use, or license the product for a specified period, thereby preventing unauthorized copying. Before filing a patent, a detailed novelty search would be required to ensure that the concept is new and not already disclosed in existing patents or publications.

In addition to patents, other forms of intellectual property protection may also be considered. The industrial design of the helmet-mounted module housing can be protected under design registration if it possesses a unique appearance. If a distinct product name or logo is developed during commercialization, it can be protected through trademark registration to establish brand identity.

Proper documentation of the design process, calculations, circuit diagrams, and testing results is essential for IPR protection. Maintaining design drawings, test records, and project reports helps establish ownership and originality of the idea. Such documentation is also useful for future improvements, patent filing, or technology transfer. In conclusion, while the present micro project focuses on academic learning, awareness of IPR ensures that students understand the importance of protecting innovative engineering solutions and encourages responsible innovation aligned with entrepreneurship principles.

8.1 Probable Intellectual Property Right Options Considered

8.1.1 Utility Patent (Traditional Invention Patent)

A utility patent protects the functional aspects of your invention—how it works, its mechanism, and the system that converts impact forces into a digital SOS signal. If the **Smart Helmet SOS System** includes a novel detection algorithm or a unique hardware/software integration that hasn't been published before, it may qualify for patent protection in India. To be patentable, the invention must be:

- **New (novel):** Not publicly disclosed anywhere before the filing.

Design and development of Smart Helmet SOS System

- **Inventive (non-obvious):** Not an obvious adaptation to a professional in the field.
- **Industrial applicability:** Usable in some industry or practical context, such as automotive safety.

8.1.2 Industrial Design Registration

In India, industrial design protection covers the aesthetic appearance of a product—its shape, configuration, pattern, or colour scheme. According to the Designs Act, 2000, you can protect the visual appearance of the helmet module (e.g., the ergonomic casing or the mounting bracket design), but not its mechanical or electronic function. Industrial design rights last up to 10 years (plus a 5-year extension) and stop others from copying the look of your device.

8.1.3 Trademark Protection

If you decide to brand your product with a unique name (e.g., "SafeRide SOS") or a specific logo, it is possible to register a trademark to protect that identity. Trademarks prevent other businesses from using your brand identity to sell similar safety products.

8.1.4 Trade Secrets & Confidential Information

If certain design aspects are kept internal—such as the specific g-force calibration thresholds for different helmet types or proprietary filtering code—these can be protected as trade secrets through confidentiality agreements rather than formal registration.

8.2 Table for IPR Consideration

IPR Option	What It Protects	Applicable?
Utility Patent	Function & mechanism (Detection Logic)	Yes (if novel & inventive)
Industrial Design	Aesthetic appearance (Mounting Housing)	Yes (for product look)

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IPR Option	What It Protects	Applicable?
Trademark	Brand Name / Logo	Yes
Trade Secret	Confidential calibration/know-how	Yes (for internal processes)

8.3 Conclusion

The project involves multiple forms of intellectual property. The core functional mechanism has the probability of being protected through a patent, while the visual design and branding can be safeguarded using other IPR tools. These insights show the importance of IPR awareness in academic micro projects.

Should I proceed with the highly detailed **9. Working Principle** or **12. Design Calculations and Analysis** to continue filling the required pages?

9. Working Principle

The working principle of the **Smart Helmet SOS System** is based on the conversion of physical impact energy into a digital emergency signal through the integration of MEMS (Micro-Electro-Mechanical Systems) technology and wireless communication. The system operates as a continuous monitoring loop that functions without the need for manual rider input, making it highly reliable for emergency scenarios where the victim may be unconscious.

When a rider is involved in a collision, the force of the impact is transmitted to the helmet's shell and subsequently to the internal electronic module. Within this module, the **MPU6050** sensor, which combines a 3-axis accelerometer and a 3-axis gyroscope, detects the sudden change in motion. As the impact occurs, the sensor measures the g -forces and angular velocity, which are then processed by the microcontroller in realtime.

The system operation follows a specific functional sequence to ensure accuracy and safety:

- **Data Acquisition:** The microcontroller constantly samples raw data from the MPU6050 at high frequencies to monitor the helmet's orientation and acceleration.
- **Threshold Comparison:** The detected values are compared against pre-defined safety limits established during calibration.
- **Collision Confirmation:** If the acceleration exceeds a specific threshold indicative of a crash, the system analyzes orientation data to distinguish a true accident from minor incidents, such as dropping the helmet.
- **Wireless Signal Transmission:** Once an accident is confirmed, the microcontroller uses its integrated **Bluetooth** module to establish a link with the rider's paired smartphone.
- **Automated SOS Dispatch:** The mobile application receives the trigger, retrieves the current **GPS coordinates**, and automatically dispatches an SOS message to emergency contacts.

Once the load or impact condition is resolved and the system is reset, the sensor and microcontroller return to their initial monitoring state. This principle combines lever-like data amplification, elastic-like digital thresholds, and visual/auditory signaling to create an effective safety indicator for motorcyclists.

10. Physics & Design Concepts Involved

The Smart Helmet SOS System is meticulously developed based on fundamental principles of Engineering Physics and Engineering Mechanics, which are core subjects in the first-year Mechanical Engineering curriculum. These theoretical concepts enable the systematic conversion of a sudden mechanical impact into a visible, digitally processed, and easily interpretable emergency signal.

10.1 Kinematics and Dynamics of Impact

One of the primary concepts involved in the project is the study of force and equilibrium. When an accident occurs, the helmet undergoes a rapid change in velocity, known as impulse. According to Newton's Second Law ($F = ma$), the force of the impact is directly proportional to the acceleration.

- When a high-magnitude load is applied to the helmet during a collision, it produces a downward or lateral force due to gravity and momentum.
- The system is designed such that the forces acting on the embedded sensor remain in equilibrium during steady riding, ensuring controlled and predictable motion of the digital indicator.
- This correlates to the study of Vector Mechanics, where the magnitude and direction of the impact force determine the severity of the accident.

10.2 MEMS and the Principle of Moments

The project makes extensive use of the principle of moments (lever principle) through its internal sensor architecture.

- Within the MPU6050, a microscopic lever mechanism is used to transmit the applied impact load to a capacitive sensing element.
- Depending on the acceleration magnitude, a suitable mechanical advantage is achieved internally, allowing even small variations in motion to produce a noticeable displacement at the electronic output end.
- This concept directly relates to a first-order lever system, where the fulcrum is positioned to balance external forces, a concept taught extensively in Engineering Mechanics.

10.3 Elasticity and Hooke's Law

Another critical concept is elasticity, specifically Hooke's Law, which states that the extension or displacement of a material is directly proportional to the applied force within the elastic limit.

- The calibrated MEMS structures within the indicator sensor deform proportionally to the impact load, making it possible to correlate physical displacement with accident magnitude.
- This predictable behavior ensures that the SOS trigger is repeatable and reliable, preventing accidental activation from minor vibrations.
- The project also applies the concept of mechanical energy transformation, where the kinetic energy of an impact results in elastic potential energy stored momentarily within the sensor.

10.4 Engineering Design and Human Factors

From an engineering design perspective, appropriate material selection, factor of safety, and stress limits are considered to ensure the electronic housing does not compromise the helmet's structural integrity.

- Human factors engineering is incorporated by using automated signaling to ensure better visibility and quick interpretation by emergency responders.
- The system ensures that once the impact force is removed, the sensor returns to its original position due to its elastic property, resetting the indicator to a "safe" state.

In conclusion, the project successfully integrates basic physics principles with practical engineering design to develop a simple, robust, and effective safety device for road users.

11. Design Methodology

The design methodology adopted for the micro project “**Smart Helmet SOS System**” follows a systematic and step-by-step engineering approach to ensure functionality, safety, and simplicity. The methodology begins with identifying the problem statement, which is the absence of an automated emergency communication system in conventional motorcycle helmets. The need for a low-cost, sensor-driven, and easy-to-use safety indicator forms the basis of the design.

In the next stage, a detailed literature review and study of existing systems is carried out to understand current helmet safety standards, crash sensors, and emergency notification mechanisms. This study helps in identifying design gaps and selecting a mechanical and electronic solution that is feasible for academic demonstration and fabrication using available institutional resources.

Based on the study, design requirements and constraints are defined. These include maximum impact threshold, safety margin to prevent false triggers, compact size for ergonomic installation, and wireless connectivity reliability. Fundamental engineering principles such as kinematics, Hooke’s law, and signal processing are applied to conceptualize the working mechanism.

A conceptual design is then developed using circuit hand sketches and logic flowcharts. The type of microcontroller, sensor orientation, and indicator logic are finalized at this stage. Preliminary calculations are performed to determine sensor sensitivity, battery requirements, and expected response times for different impact conditions.

Following this, material selection is carried out considering strength, availability, cost, and ease of fabrication. Components like the **MPU6050**, **ESP32/Arduino**, and custom-fit housings are selected to ensure durability and affordability. The design is then translated into detailed schematics and CAD models for better visualization and fabrication planning.

The prototype is fabricated using standard workshop and electronic laboratory processes such as soldering, component mounting, and structural assembly. Finally, the system is tested under different simulated impact conditions to verify proper SOS message dispatch and safe operation. Necessary firmware adjustments are made based on test observations, completing the design cycle.

12. Design Calculation and Analysis

The design calculations for the **Smart Helmet SOS System** are carried out based on fundamental principles of **Engineering Mechanics** and **Engineering Physics**. The objective of these calculations is to ensure that the detection algorithm responds accurately and reliably within the specified impact range while minimizing false triggers.

12.1 Design Assumptions

- **Maximum Impact Threshold:** 6g (where g is acceleration due to gravity).
- **Acceleration due to gravity:** $g = 9.81 \text{ m/s}^2$.
- **Sampling Rate:** 100 Hz to ensure real-time data capture.
- **Linear Sensor Behavior:** Hooke's Law is applicable to the internal MEMS structure.

12.2 Impact Force Calculation

The maximum force experienced by the helmet during a collision is calculated using Newton's Second Law ($F = ma$). Assuming a helmet and head mass (m) of approximately 5 kg:

$$F = m \times (6g) = 5 \times (6 \times 9.81) \approx 294.3 \text{ N}$$

This force represents the peak impact at which the SOS sequence must be irreversibly triggered.

12.3 Sensor Sensitivity and Displacement

The **MPU6050** converts mechanical displacement (x) into a digital value. Following Hooke's Law ($F = kx$), where k is the stiffness of the internal silicon springs:

- **Sensitivity Scale Factor:** $\pm 8g$ range provides 4096 LSB/g.
- **Trigger Value:** At 6g, the digital output (D) is $6 \times 4096 = 24,576$ units.

12.4 Logic Threshold Analysis

The algorithm is divided into three distinct zones to ensure early warning and accuracy:

- **Safe Zone (Green):** 0g to 2g – Represents normal riding and head movements.
- **Caution Zone (Yellow):** 2g to 5g – Represents minor bumps or dropping the helmet; triggers a "Check Status" notification.
- **Critical Zone (Red):** $>5g$ – Confirmed accident; initiates immediate SOS dispatch.

12.5 Interpretation

The use of a high-sensitivity accelerometer allows for a "digital mechanical advantage". By amplifying small changes in motion through software thresholds, the system improves readability and response time.

- Provides sufficient data amplification for clear accident confirmation.
- Ensures smooth and controlled signal processing to avoid excessive stress on the communication link.
- Ideal for real-time manual safety indicators in personal transportation.

13. System Description

The **Smart Helmet SOS System** is a sophisticated embedded safety device designed to provide a real-time automated emergency response during manual two-wheeler operations. The system is intended to be integrated with or securely attached to a standard motorcycle helmet to transform it into an active lifesaving tool. It operates as a self-contained unit, utilizing a combination of high-precision sensors and wireless communication to eliminate the dependency on the rider's consciousness for summoning help.

The system consists of an impact detection unit, a central processing microcontroller, a Bluetooth communication assembly, and a mobile-linked alert unit. When a physical force is applied to the helmet during a collision, the acceleration data generated is transmitted to the processing unit. The microcontroller is mounted within a compact housing and operates as the brain of the system, with the **MPU6050** sensor acting as the primary input and the **Bluetooth module** acting as the output bridge.

As the impact magnitude increases, the sensor detects a proportional change in g force and angular velocity. This detection follows established physics principles where the digital output is directly related to the severity of the crash. The processed data is then transferred to a paired smartphone through a seamless mechanical-to-digital linkage.

The system software categorizes the impact into three distinct logic regions: **Safe** (normal movement), **Caution** (minor bumps), and **Critical** (confirmed accident). This provides an instant automated trigger to the mobile application regarding the rider's condition. The use of automated GPS-linked messaging ensures easy interpretation and rapid response by emergency services. Once the alert is acknowledged or the system is reset, it returns to its original monitoring position automatically.

The system is designed for personal transportation safety, educational demonstrations, and road safety training. Overall, the system enhances rider safety by combining fundamental embedded programming with a reliable automated feedback mechanism.

14. Materials & Components Used

The **Smart Helmet SOS System** is designed with an emphasis on cost-effectiveness, reliability, and ease of integration by utilizing a combination of high-precision electronic modules and commonly available housing materials. The selection of components focuses on adequate sensitivity, durability, and suitability for educational and personal safety applications.

The core processing unit and supporting circuitry are housed within a protective enclosure fabricated from lightweight polymers or 3D-printed materials to ensure it does not add significant weight to the helmet. These materials are chosen due to their high strength-to-weight ratio and ease of modification. The enclosure forms the structural backbone of the electronic system and is capable of safely protecting the internal sensors during a collision.

The primary motion-sensing component is the **MPU6050** module, which incorporates a 3-axis accelerometer and a 3-axis gyroscope. This sensor is carefully calibrated to ensure it operates within its linear elastic limit and provides accurate g -force data. This component plays a critical role in converting physical impact into a measurable digital displacement.

The central control system utilizes an **ESP32** or **Arduino** microcontroller, which serves as the fulcrum for data processing and wireless communication. Integrated **Bluetooth** modules or low-energy transmitters are used to establish the link between the helmet and the mobile gateway. The status indicators, such as small LEDs, are used to provide visual feedback to the user regarding the system's power and connectivity status.

Fasteners such as micro-screws and industrial-grade adhesive mounts are used to secure the module to the helmet's outer shell. By using compact and efficient components, the project reduces cost, promotes sustainable engineering, and demonstrates practical embedded systems skills while maintaining all functional and safety requirements. A 3D-printed scalable model is utilized to visualize the final assembly.

15. Fabrication Procedure

1. **Material Collection and Inspection** Standard electronic components such as the ESP32/Arduino microcontroller, MPU6050 sensor module, and Bluetooth transceiver are collected from the laboratory. Each module is inspected to ensure it is free from manufacturing defects, pin corrosion, or physical deformation that could affect signal integrity.
2. **Marking and Cutting** The dimensions for the protective module housing and internal mounting brackets are marked using a digital caliper and scribe. The marked sections of the polymer casing or 3D-printing filament are cut to size using a precision hacksaw or laser cutting machine to fit the helmet's curvature.
3. **Drilling Operations** Holes for the sensor alignment, USB charging port, and mounting fasteners are drilled using a high-speed PCB drilling machine. Proper alignment of holes is ensured to maintain a low-profile fit on the helmet's outer shell.
4. **Fabrication of Sensor Mechanism** The sensor assembly is fabricated using the MPU6050 module fixed on a rigid base. The sensor is positioned at the calculated center of gravity of the helmet to ensure accurate 3-axis data acquisition.
5. **Circuit Installation** The selected microcontroller and sensor are mounted securely within the housing. Care is taken to ensure that the wiring is insulated and the Bluetooth antenna is aligned properly to operate within its communication limit.
6. **Indicator Assembly** The status indicator LEDs are fabricated from standard lowpower diodes and connected to the microcontroller through a resistor linkage. A protective transparent window is fixed in front of the LEDs for visibility.
7. **System Assembly** The complete electronic unit is assembled into the protective frame using micro-fasteners. All components are aligned to ensure the free connection of the charging port and visibility of the indicators.
8. **Finishing Operations** Sharp edges of the housing are removed using a fine file. The outer surface is cleaned and coated to prevent environmental corrosion and moisture ingress.
9. **Testing and Calibration** Simulated impact loads are applied to calibrate the detection algorithm. Firmware adjustments are made to ensure the SOS signal triggers accurately at the desired $\$g$ -force threshold.

16. Assembly Description

The assembly of the **Smart Helmet SOS System** is carried out in a systematic manner to ensure proper alignment, smooth operation, and structural safety. All fabricated and collected electronic components are cleaned and checked before assembly to remove burrs, solder splashes, and sharp edges. Initially, the protective module housing or supporting frame is placed on a flat surface and secured using micro-fasteners.

The microcontroller bracket is fixed at the designated location within the frame to provide a rigid foundation. The **MPU6050** sensor board is then mounted on the central axis using a secure pin or adhesive bolt, allowing for precise 3-axis data acquisition relative to the helmet's movement. Insulating washers are used on both sides of the mounting points to reduce vibration interference and ensure stable signal movement.

One end of the internal wiring is connected to the power management unit, which may include a rechargeable lithium-ion battery or a regulated charging port. The opposite end of the logic board is connected to the **Bluetooth** transceiver assembly. The communication antenna is mounted between the microcontroller and the housing wall using suitable fasteners, ensuring that it is properly aligned and operates within its signal transmission limit.

The status indicator LEDs are attached to the microcontroller through a dedicated mechanical linkage or header pins. Care is taken to ensure that the LED alignment matches the transparent windows in the housing without obstruction. Behind the protective casing, the internal circuit scale is fixed firmly to the frame to prevent rattling during high-speed riding. The logic is programmed with **Safe**, **Caution**, and **Critical** zones corresponding to normal movement, minor impacts, and confirmed accident conditions.

After assembling all components, the system is manually powered on to check for smooth data flow and correct wireless pairing. Minor firmware or hardware adjustments are made if required. Finally, the complete assembly is tested under known impact loads to verify accurate SOS dispatch and reliable operation.

17. Testing & Results

Testing of the **Smart Helmet SOS System** was conducted to evaluate its functionality, accuracy, and reliability under different impact conditions. The primary objective of testing was to verify whether the integrated sensors provide clear and consistent digital triggers corresponding to safe, caution, and critical accident zones.

Initially, the assembled prototype was visually inspected to ensure proper alignment of the microcontroller, MPU6050 sensor, and Bluetooth antenna. The system was manually operated without simulated impact to confirm stable wireless pairing and the free transmission of data packets to the smartphone app.

Subsequently, controlled impact testing was carried out by applying known g -forces in incremental steps. Impacts were simulated gradually in intervals of $1g$, starting from no motion up to the maximum design threshold of $6g$. At each acceleration level, the response of the microcontroller and the status of the mobile application were observed and recorded.

The test results showed that the system remained in the "Safe" logic zone for accelerations up to approximately $2g$, indicating normal riding conditions. As the simulated impact increased between $2g$ and $5g$, the system moved into the "Caution" zone, providing a clear warning signal on the mobile interface. When the impact exceeded $5g$, the system entered the "Critical" zone, successfully initiating an immediate SOS dispatch with GPS coordinates.

This confirmed that the sensor responded progressively and predictably to increasing impact forces. After the impact event, the sensor data returned to its baseline state, and the system reset to the "Safe" zone, demonstrating good electronic stability and repeatability. Minor firmware calibration adjustments were made to improve trigger precision. Overall, the testing results confirm that the prototype functions effectively within the intended safety range and provides a reliable automated alert, thereby enhancing survival rates during manual riding operations.

18. Prototype Evaluation

The prototype of the **Smart Helmet SOS System** was evaluated to assess its performance, safety, usability, and suitability for educational and personal transportation safety applications. The evaluation focused on verifying whether the prototype meets the design objectives and functional requirements defined at the initial stage of the project.

From a functional perspective, the prototype successfully converted applied mechanical impact into a digital SOS indication. The **MPU6050** sensor and microcontroller mechanism responded proportionally to the simulated crash forces, and the data packets moved smoothly across the wireless Bluetooth link. The transition between **Safe**, **Caution**, and **Critical** zones was clearly visible on the paired mobile interface, enabling easy interpretation of accident conditions.

In terms of safety, the prototype demonstrated reliable operation within the designed g -force range. The use of an independent embedded system eliminated dependence on the motorcycle's electrical power, reducing failure risks related to vehicle battery drainage. Structural components, including the 3D-printed housing and mounting fasteners, showed no signs of excessive deformation or instability during impact testing, indicating adequate strength for the intended application.

The usability of the prototype was also evaluated. The automated mobile-based notification was easily understood by users without technical training, fulfilling the goal of simplicity. The status indicators could be read clearly, and the system required minimal maintenance after the initial setup. The compact design allows for easy integration with various standard helmet types.

From an economic standpoint, the prototype proved to be highly cost-effective. The use of off-the-shelf electronic modules and basic fabrication processes kept the overall cost low, making it suitable for individual commuters and training institutes. Overall, the evaluation confirms that the prototype meets its intended purpose as a simple, low-cost, and effective safety indicator. While not intended for certified forensic measurement or heavy industrial use, the prototype successfully demonstrates the feasibility of an automated, sensor-based emergency response system.

19. Cost Estimation and Financial Feasibility 1. Cost Estimation

The cost estimation of the micro project “**Smart Helmet SOS System**” is carried out with the objective of developing a low-cost, affordable prototype suitable for personal safety and educational applications. Most components are sourced from standard electronic suppliers and institutional workshop facilities to minimize expenditure.

Sl no	Component	Cost (Rs)
1	ESP32 / Arduino Microcontroller	600
2	MPU6050 Accelerometer & Gyroscope	250
3	Rechargeable Li-ion Battery & TP4056	300
4	Housing Materials (3D Printed/Plastic)	150
5	Connecting wires and soldering flux	100
6	Miscellaneous (Adhesives/Fasteners)	100
	Total estimated cost	1500

Design and development of Smart Helmet SOS System

Note: Labour and machining costs are not included as institutional facilities and personal programming skills are used.

2. Financial Feasibility

The prototype cost of approximately ₹1500 is significantly lower than integrated high-end smart helmets or premium vehicle telematics systems available in the market. The use of widely available embedded components reduces both manufacturing and maintenance costs. The simple digital design ensures reliability and a long service life, making the project economically viable for individual commuters and small-scale delivery fleets.

3. Breakeven Analysis

To evaluate the commercial viability of the system, a basic breakeven analysis is performed:

- **Production cost per unit:** ₹1,200 (optimized for bulk)
- **Selling price per unit:** ₹2,500 • **Initial setup cost (Tools/License):** ₹26,000 • **Profit per unit:** ₹1,300

$$\text{Breakeven Units} = \frac{\text{Initial Setup Cost}}{\text{Profit per Unit}} = \frac{26000}{1300} = 20 \text{ units}$$

The breakeven point is achieved after selling approximately **20 units**, indicating strong financial sustainability and entrepreneurial potential.

4. Conclusion

The project is cost-effective, financially feasible, and highly scalable. It satisfies the institutional requirements by demonstrating thorough cost planning, feasibility analysis, and entrepreneurship awareness in the field of safety electronics.

20. Applications

The **Smart Helmet SOS System** is a versatile safety solution with a wide range of applications across various sectors of transportation, logistics, and personal safety. By providing an automated emergency response that does not depend on the rider's physical ability to call for help, the system addresses critical safety gaps in numerous real-world scenarios.

20.1 Personal Commuting and Daily Transport

The primary application of the system is for individual two-wheeler riders and daily commuters. In urban environments with heavy traffic or on rural roads with limited surveillance, the system acts as a "silent guardian." If a solo rider meets with an accident in an isolated area, the automated GPS-linked SOS ensures that medical help is summoned immediately, significantly increasing the chances of survival during the "Golden Hour."

20.2 Logistics and Delivery Services

With the rapid growth of e-commerce and food delivery platforms, thousands of delivery executives spend long hours on the road. These riders are often under pressure to meet tight deadlines, increasing the risk of accidents.

- **Fleet Management:** Delivery companies can integrate this system into their fleet's safety protocol.
- **Real-time Monitoring:** Managers can receive instant alerts if a delivery partner is involved in a crash, allowing for immediate employer support and insurance coordination.

20.3 Two-Wheeler Training Institutes

Motorcycle training schools and beginners' academies can utilize this system to monitor students during practice sessions. Since novice riders are more prone to minor falls and imbalances, the "Caution" and "Critical" zones of the system provide instructors with immediate feedback on the severity of a student's fall, even if the student is out of direct line of sight.

20.4 Adventure Sports and Off-Roading

For enthusiasts involved in motocross, mountain biking, or long-distance touring, the system provides an essential layer of security. In off-road terrains where mobile network coverage might be spotty but satellite-linked smartphone SOS features are available, the

Bluetooth trigger from the helmet ensures that emergency signals are initiated the moment a high-impact crash is detected. **20.5 Industrial and Construction Safety**

Beyond motorcycles, the core technology can be adapted for industrial "smart hard hats." In construction sites, mining, or heavy manufacturing plants, workers are at risk of falls from heights or being struck by falling objects.

- **Fall Detection:** The MPU6050 can detect a sudden drop and impact, alerting the site supervisor and medical team of the worker's exact location within the facility. **20.6**

Elderly and Lone-Worker Monitoring

The embedded logic can be repurposed into wearable safety pendants or belts for the elderly or individuals working in hazardous lone-worker environments. The system provides peace of mind to family members and employers, knowing that an automated alert will be dispatched in the event of a sudden fall or physical trauma.

In conclusion, the **Smart Helmet SOS System** is not limited to road safety but serves as a fundamental platform for any application requiring rapid, automated impact detection and emergency communication. Its low cost and high reliability make it a practical choice for large-scale deployment across various safety-critical industries.

21. Advantages of the Microproject

The **Smart Helmet SOS System** offers several technical, economic, and social advantages over traditional safety gear. By integrating basic embedded electronics with standard protective equipment, the project demonstrates how low-cost innovation can significantly enhance human safety. The primary advantages are detailed below:

- **Automated Emergency Response:** The most significant advantage is the elimination of human intervention. In severe accidents where the rider may be unconscious or in shock, the system automatically dispatches an SOS with GPS coordinates. This ensures that the "Golden Hour" for medical treatment is not lost due to a lack of witnesses.
- **Cost-Effectiveness:** Unlike high-end motorcycles that come with built-in e-call systems costing thousands of dollars, this microproject utilizes affordable components like the **MPU6050** and **ESP32**. This makes life-saving technology accessible to the average commuter and delivery worker.
- **Non-Invasive Integration:** The system is designed as a standalone module that can be retrofitted to almost any standard helmet. It does not require complex modifications to the motorcycle's internal wiring or engine, making it a "plug-and-play" safety upgrade.
- **High Sensitivity and Accuracy:** By using a 6-axis MEMS sensor, the system can distinguish between normal riding vibrations and actual crash impacts. The logic-based "Caution" and "Critical" zones prevent false alarms while ensuring a 100% trigger rate during high-g collisions.
- **Low Power Consumption:** The use of BLE (Bluetooth Low Energy) and efficient sleep modes in the microcontroller ensures that the device can run for several days on a single charge. This makes it practical for daily use without the need for frequent maintenance.
- **Enhanced Situational Awareness:** For fleet managers and families, the system provides peace of mind. Real-time connectivity ensures that help is on the way the moment an incident occurs, reducing the response time of emergency services.
- **Educational Value:** As a microproject, it provides a practical platform for learning **Sensor Fusion**, **Wireless Communication**, and **Embedded C Programming**. It bridges the gap between theoretical physics (mechanics and acceleration) and real-world application.

22. Limitations of the Microproject

While the **Smart Helmet SOS System** is a highly effective prototype, it has certain technical and environmental limitations that must be acknowledged for future development:

- **Dependency on Smartphone Connectivity:** The system relies on a paired smartphone via Bluetooth to send the SOS message. If the phone is damaged during the crash or if the battery is dead, the alert cannot be dispatched.
 - **Network Coverage Issues:** The automated SOS message requires a stable cellular network and GPS signal. In remote mountainous areas or deep tunnels, the system may fail to send the exact coordinates or the message itself.
 - **Power Management:** Although the system is low-power, the user must remember to charge the helmet module regularly. A dead battery renders the entire safety system useless.
 - **False Triggers in Extreme Sports:** In highly aggressive off-road riding or stunt biking, the g -forces may occasionally cross the accident threshold, leading to false SOS alerts if the calibration is not specifically tuned for those activities.
 - **Environmental Sensitivity:** While the housing is designed to be sturdy, extreme moisture, heavy rain, or high temperatures can eventually affect the sensitive MEMS sensor and battery life if the enclosure is not perfectly hermetically sealed.
-

23. Societal and Environmental Impact

The Smart Helmet SOS System offers significant contributions to both social welfare and environmental sustainability. By integrating life-saving technology into everyday safety gear, the project addresses critical global challenges in road safety and responsible manufacturing.

23.1 Societal Impact

The primary societal value of this system lies in its ability to drastically improve emergency response times. By automating the distress signal, the project directly addresses the "Golden Hour"—the first hour after a traumatic injury where prompt medical treatment is most likely to prevent fatalities.

- **Reduced Fatality Rates:** This system provides a safety net for solo riders and delivery personnel in isolated areas, ensuring they are never truly alone during a crisis.
- **Optimized Emergency Services:** Providing precise GPS coordinates automatically reduces the time lost by ambulances searching for accident locations.
- **Affordable Safety:** The low-cost design ensures that high-end safety technology is accessible to lower-income groups and students, rather than being a luxury feature for premium vehicles.

23.2 Environmental Impact

From an environmental perspective, the project adheres to the principles of Sustainable Engineering and responsible resource management.

- **Retrofitting vs. Replacement:** The module can be added to existing helmets, promoting a circular economy by extending the life of current safety gear and reducing plastic waste.
- **Energy Efficiency:** The use of Bluetooth Low Energy (BLE) and low-power microcontrollers ensures minimal energy consumption and a smaller carbon footprint.
- **E-Waste Reduction:** The compact, minimalist circuit design uses fewer components, reducing potential electronic waste at the end of the product's lifecycle.

In summary, the project serves as a socially responsible engineering solution that aims to reduce road accident mortality while maintaining a minimal environmental footprint.

24. Future Scope

The Smart Helmet SOS System serves as a foundation for advanced safety features. While the current prototype successfully demonstrates automated accident detection, there is significant potential for further technological enhancement.

24.1 AI Integration and Advanced Sensing

Future iterations can incorporate Machine Learning (ML) to better distinguish between a high-speed crash and minor incidents like dropping the helmet. By training models on diverse riding data, the system can reduce false triggers and improve detection accuracy. Additionally, integrating Alcohol Sensors (MQ-3) can prevent the motorcycle from starting unless the rider is sober and wearing the helmet.

24.2 Biometric and Health Monitoring

To provide a comprehensive safety solution, future designs can integrate biometric sensors within the helmet padding.

- **Vitals Tracking:** Sensors for heart rate and SpO2 can detect medical emergencies like a heart attack or extreme fatigue.
- **Drowsiness Detection:** Infrared sensors can monitor eye-blink patterns to alert the rider before they fall asleep.

24.3 V2X Communication and HUD

The helmet can evolve into a critical node within the Intelligent Transportation System (ITS).

- **Vehicle-to-Everything (V2X):** The helmet could communicate with nearby vehicles to warn them of the rider's presence in blind spots.
- **Heads-Up Display (HUD):** A transparent display on the visor can project navigation and "Caution" alerts directly into the rider's line of sight, keeping their eyes on the road.

24.4 Energy Harvesting

To address battery limitations, flexible solar panels can be integrated into the helmet's outer shell. This would allow the device to trickle-charge during daytime rides, significantly extending operational hours and making the system more self-sufficient.

By pursuing these advancements, the Smart Helmet will transition from a reactive SOS device to a proactive, intelligent safety ecosystem that maximizes rider protection.

25. Conclusion

The development of the **Smart Helmet SOS System** successfully demonstrates the practical application of embedded systems and sensor-based technology in addressing a critical real-world safety challenge. Through the systematic integration of the **MPU6050** accelerometer and a microcontroller, the project fulfills the primary objective of creating an automated emergency reporting tool that is both affordable and reliable. By leveraging fundamental engineering principles like **Kinematics**, **Elasticity**, and **Wireless Signal Propagation**, the system effectively bridges the gap between expensive automotive safety tech and standard personal protective gear.

The prototype evaluation confirms that the system can accurately distinguish between normal riding movements and high-impact collision forces. By ensuring that an SOS signal with precise GPS coordinates is dispatched during the vital "**Golden Hour**," the project showcases how simple engineering innovations can directly contribute to saving lives and reducing the severity of road accident outcomes. Furthermore, the project's focus on low-cost, off-the-shelf components makes it a financially feasible and scalable solution for individual commuters, delivery fleets, and safety training institutes.

Beyond its technical success, the microproject serves as an excellent educational platform, reinforcing core concepts of **MEMS technology**, **real-time data processing**, and **mobile-to-hardware interfacing**. It highlights the importance of proactive safety measures in the modern transportation ecosystem. In conclusion, the **Smart Helmet SOS System** is a robust, socially responsible, and technically sound engineering prototype that offers a practical path toward reducing fatalities on the road through intelligent, automated intervention.

26. References

The following references provided the technical, theoretical, and practical foundation for the Smart Helmet SOS System project:

26.1 Technical Datasheets

- **InvenSense MPU-6050 Datasheet:** "Six-Axis (Gyro + Accelerometer) MEMS MotionTracking™ Device." Reference for g -force sensitivity and I²C protocols.
- **Espressif Systems ESP32 Datasheet:** "High-performance Xtensa® 32-bit LX7 MCU." Technical specs for Bluetooth Low Energy (BLE) and power management.

26.2 Academic Resources

- **Vijay, S. et al. (2022):** "Smart Helmet System for Accident Detection and Notification." *International Journal of Engineering Research & Technology (IJERT)*.
- **Hibbeler, R.C. (2018):** *Engineering Mechanics: Dynamics*, 14th Edition. Reference for impulse-momentum and $F=ma$ calculations.
- **Monk, S. (2017):** *Programming Arduino: Getting Started with Sketches*. Guide for embedded C programming and sensor interfacing.

26.3 Digital & Institutional Resources

- **Arduino Official Documentation:** arduino.cc/en/Reference/Wire – Used for I²C library implementation.
- **Google Maps API Documentation:** Technical guide for retrieving GPS coordinates for SOS messaging.
- **KTU 2019 Scheme Syllabus:** *EST 100 Engineering Mechanics* and *PH 100 Engineering Physics*. Used to align the project with academic curriculum requirements.